

Descriptive Analytics Report: Amazon Product Sales Performance

Best-Sellers, Revenue Trends, Regional Demand & Pricing Patterns

Note on data: this report analyzes an Amazon-style order-level dataset of **60 transactions** collected over a **one-week period**, spanning **8 products** across **2 categories** (Accessories, Electronics) and **4 regions** (North, South, East, West). Each row captures one order: product, unit price, quantity, region, and order total. Prices are treated as Indian Rupees (Rs). This structure mirrors what a descriptive analytics layer over checkout data would surface at any scale.

Prepared for	Amazon Sales Performance Review
Dataset	amazon_sales_dataset.xlsx
Records analyzed	60 orders
Report type	Descriptive Analytics
Date	August 2026

Note on data :

This report analyzes 60 Amazon order-level transactions, collected over a one-week period, to build a descriptive analytics view across product, category, region, and pricing dimensions. Across the dataset, orders totaled **Rs 4,474,143** in revenue on **194** units, for an average order value of **Rs 74,569**.

Mouse emerged as the clear best-seller by units, while **Mouse** led on revenue. **North** and **South** together account for the majority of regional demand, and **Accessories** outperforms **Electronics** on both revenue and order count overall -- though this reverses in the West region. Unit price and order quantity showed essentially no correlation ($r = 0.02$), a finding with direct implications for how discounting and bundle strategies are evaluated.

1. Objectives

This report builds a descriptive analytics layer over Amazon order-level transaction data to answer ten recurring business questions that a merchandising, pricing, or regional operations team would ask on a recurring basis:

- **Objective 1:** What are the best-selling products by units sold?
 - **Objective 2:** Which products generate the most revenue?
 - **Objective 3:** How is revenue distributed across product categories?
 - **Objective 4:** Where is customer demand concentrated geographically?
 - **Objective 5:** How does average order value (AOV) vary by region?
 - **Objective 6:** How does average unit price vary across products?
 - **Objective 7:** What does the distribution of order quantities look like?
 - **Objective 8:** How does each region split its spend across categories?
 - **Objective 9:** How does order volume (not just revenue) split by region and category?
 - **Objective 10:** Is there a relationship between unit price and quantity ordered?
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How This Would Work At Real Scale

This analysis didn't need a data-engineering stack -- it was produced entirely inside two everyday AI chat apps, Claude and ChatGPT, using the raw spreadsheet as the only input. A much simpler process:

Step	Tool used
Upload the spreadsheet	Claude / ChatGPT (attached directly in the chat)
Descriptive analysis & calculations	Claude
Chart creation	Claude
Cross-checking figures	ChatGPT
Report writing & formatting	Claude

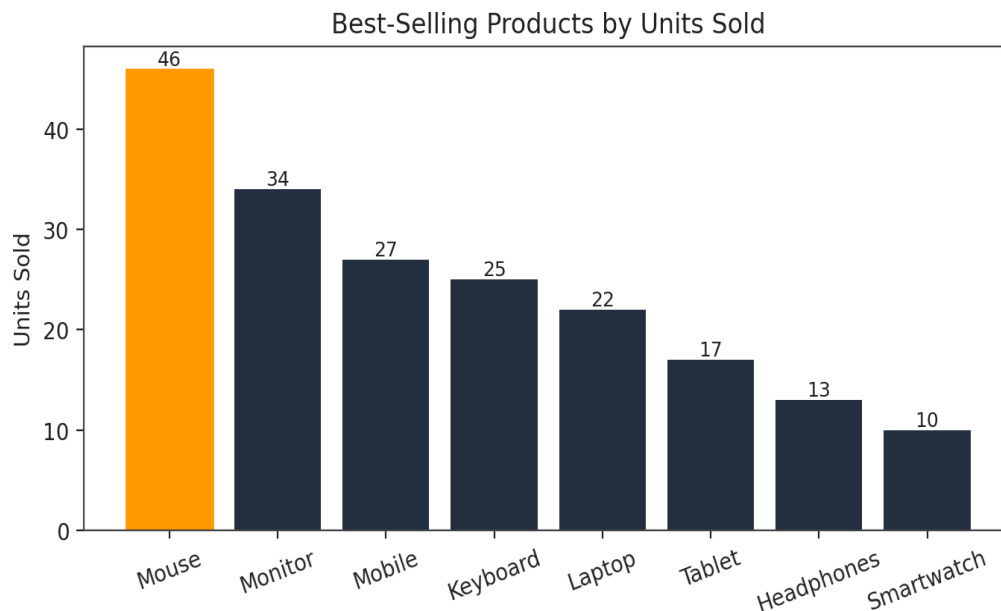
This report is built from a static 60-row extract analyzed entirely within Claude and ChatGPT conversations -- no pipeline, warehouse, or BI tool involved. If the underlying data grows or refreshes, the same charts and tables can be regenerated simply by re-uploading the updated file to either app.

Objective 1: Best-Selling Products by Units Sold

Mouse leads with 46 units sold across 8 products tracked.

Mouse is the clear volume leader at **46 units**, roughly 35% ahead of the second-place **Monitor** (34 units). This concentration is a classic "hero SKU" pattern in e-commerce, where a small number of listings drive a disproportionate share of volume -- a signal worth flagging for inventory and reorder prioritization.

Ranking logic: this view ranks by units, not revenue -- a low-priced accessory outselling a high-priced electronic matters differently to warehousing than it does to finance, so both cuts (units and revenue) are presented as separate views in this report (see Objective 2).



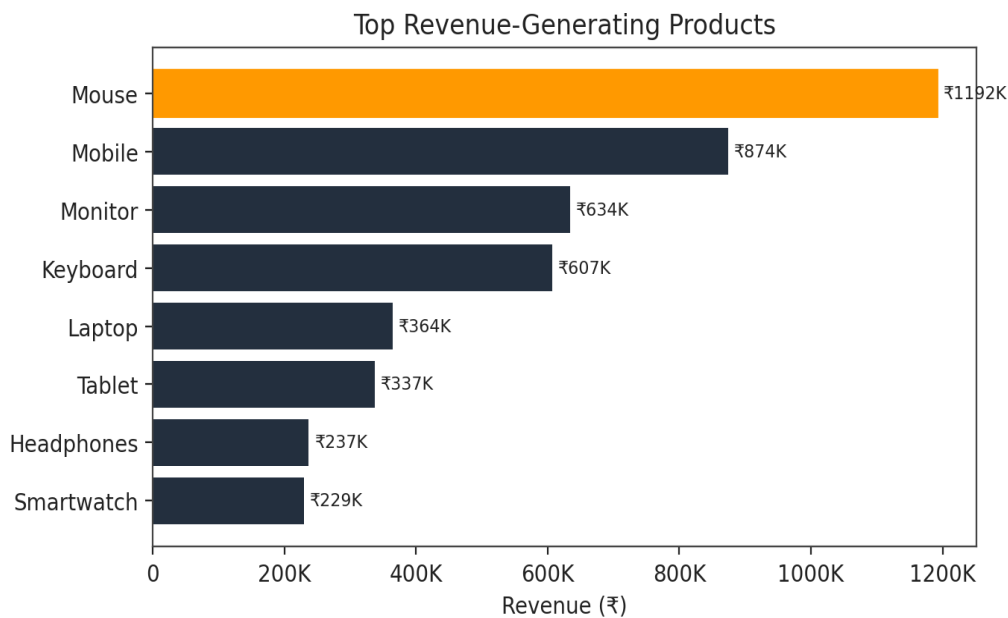
KPI — Top product by units: Mouse — 46 units

Objective 2: Top Revenue-Generating Products

Mouse generated Rs 1,191,928 in revenue -- the top earner overall.

Mouse tops the revenue leaderboard at **Rs 1,191,928**, followed by **Mobile** at Rs 874,045. Notably, the revenue ranking does not exactly mirror the units ranking from Objective 1 -- Mouse combines strong volume with a relatively high average unit price (Rs 26,591), while lower-volume products with high price points (e.g. Mobile) still punch above their unit count in revenue terms.

Why this matters: a product team optimizing for revenue and one optimizing for shelf/warehouse turnover should be looking at two different leaderboards -- this is exactly the kind of divergence a single "best-seller" metric would hide.



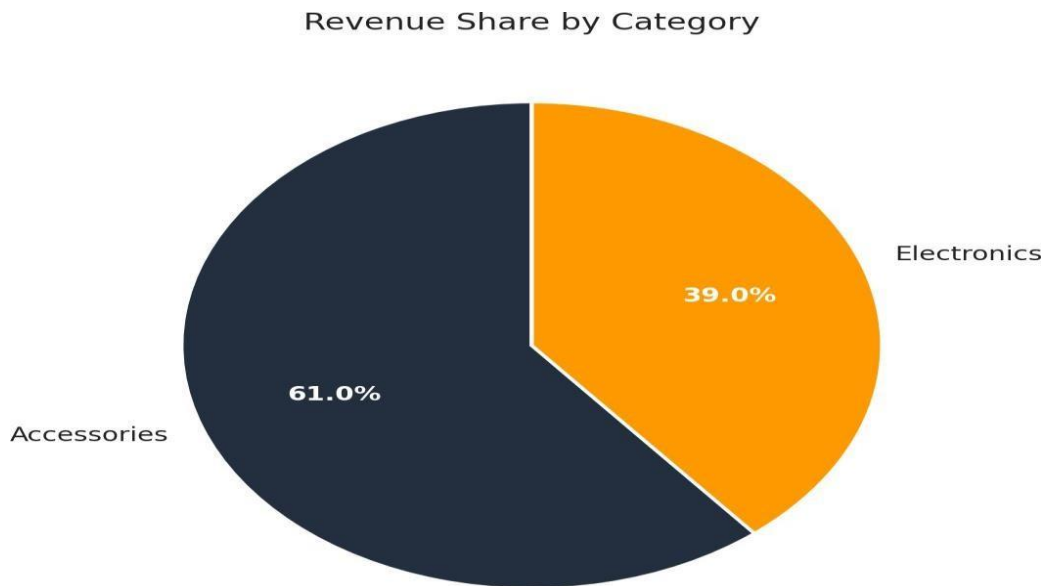
KPI — Top product by revenue: Mouse — Rs 1,191,928

Objective 3: Revenue Distribution by Category

Accessories account for 61.0% of revenue vs. 39.0% for Electronics.

Accessories generated **Rs 2,727,751** (61.0% of total revenue) across 114 units, while **Electronics** generated **Rs 1,746,392** (39.0%) across 80 units. Accessories also carries the larger share of order count (35 of 60 orders), suggesting it is both the higher-frequency and higher-revenue category in this dataset -- not simply a high-volume, low-value complement to Electronics.

In a production dashboard, this split would typically be tracked weekly to catch category-level demand shifts before they show up in aggregate revenue figures.



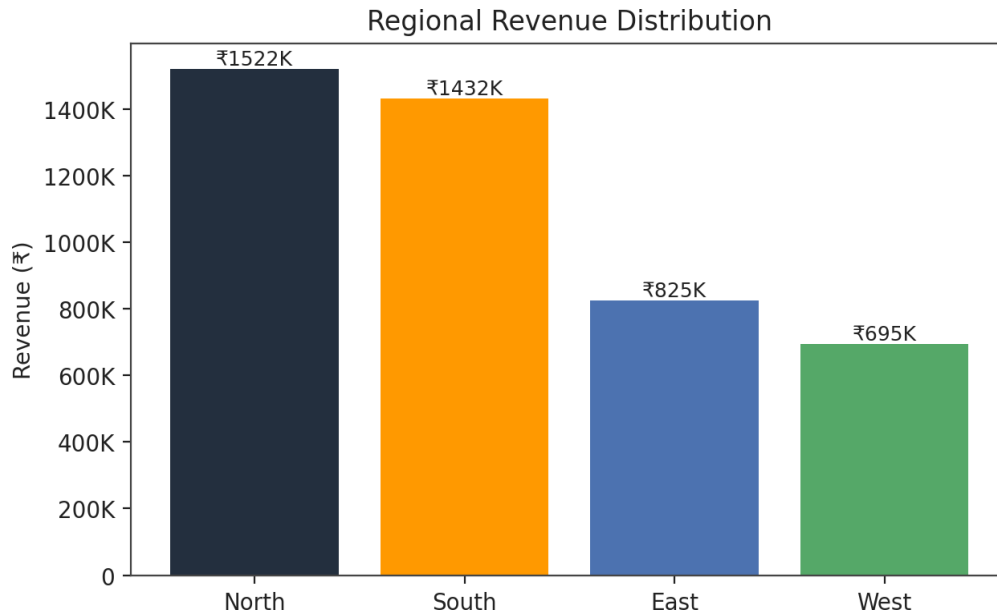
KPI — Accessories share of revenue: 61.0% (Rs 2,727,751)

Objective 4: Regional Demand Concentration

North leads all regions with Rs 1,521,630 in revenue (18 orders).

Demand is led by **North** (Rs 1,521,630, 18 orders) and **South** (Rs 1,432,154, 18 orders), together accounting for 66% of total revenue. **West** trails the field at Rs 695,438, roughly 54% below the leading region.

This view is what would drive regional warehouse and fulfillment-center allocation, localized pricing or promotion decisions, and shipping SLA risk assessment -- regions with concentrated demand and thin local inventory are the ones most exposed to stockout risk.



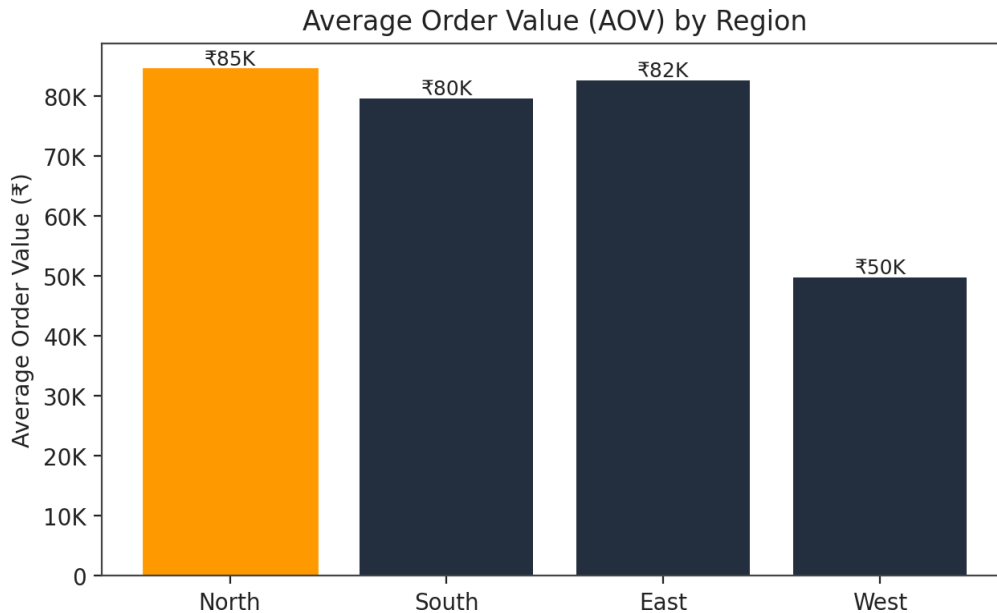
KPI — Leading region by revenue: North — Rs 1,521,630

Objective 5: Average Order Value (AOV) by Region

North has the highest AOV at Rs 84,535 per order.

North customers place the highest-value orders on average (Rs 84,535), while **West** trails at Rs 49,674 -- a gap of 41%. Comparing this to Objective 4 is instructive: West also has the lowest total revenue, and a below-average AOV compounds that gap rather than offsetting it with higher order frequency.

AOV by region is a useful input for deciding where free-shipping thresholds, bundle promotions, or minimum-order incentives would have the most leverage.



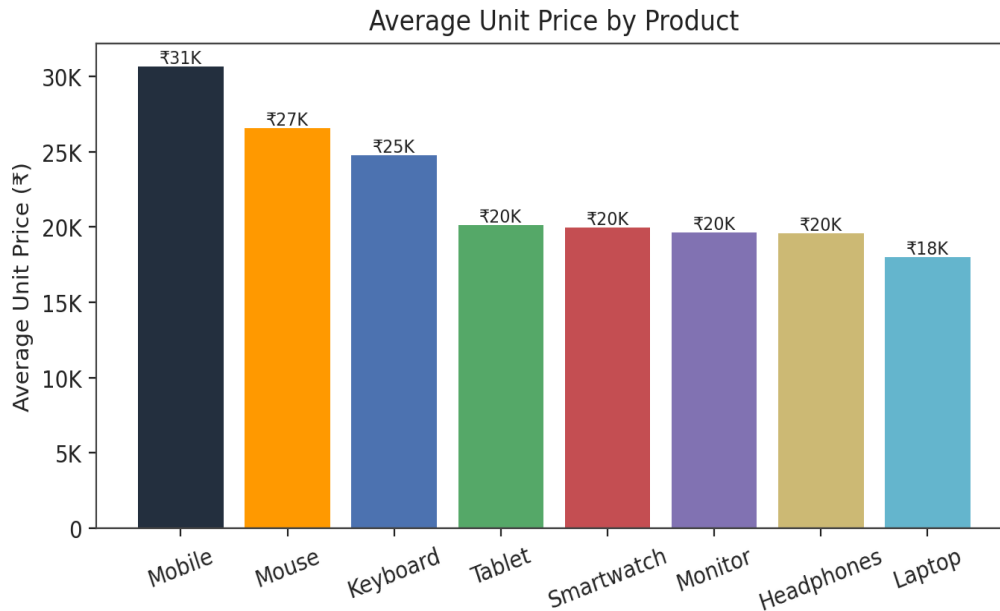
KPI — Highest AOV region: North — Rs 84,535 / order

Objective 6: Average Unit Price by Product

Mobile carries the highest average price at Rs 30,672.

Average unit price ranges from **Rs 30,672** (Mobile) down to **Rs 17,991** (Laptop) -- a 1.7x spread across the product catalog. This is unit price only, not units-weighted; it tells a pricing team where the catalog sits on the price ladder, independent of how well each product actually sells.

Read alongside Objective 1 (units) and Objective 2 (revenue), this helps separate products that sell well *because* of price from products that sell well *despite* price.



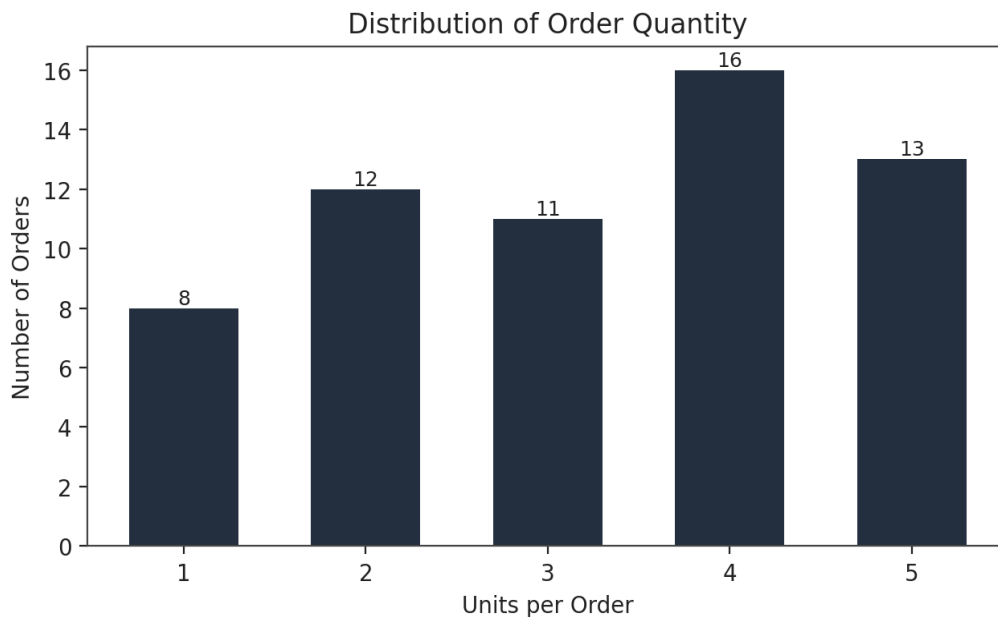
KPI — Highest average unit price: Mobile — Rs 30,672

Objective 7: Distribution of Order Quantity

Orders of 4 units are the most common, at 16 of 60 orders.

Order size is fairly evenly spread from 1 to 5 units per order, with **4-unit orders** the single most frequent bucket (16 orders, 27%). Single-unit orders are the least common (8 orders), suggesting most customers in this dataset are buying in small multiples rather than one-off single-item purchases.

In a production system, this distribution would be tracked over time to detect shifts toward bulk buying (which changes packaging and shipping-cost assumptions) or toward single-unit orders (which raises per-order fulfillment cost as a share of revenue).



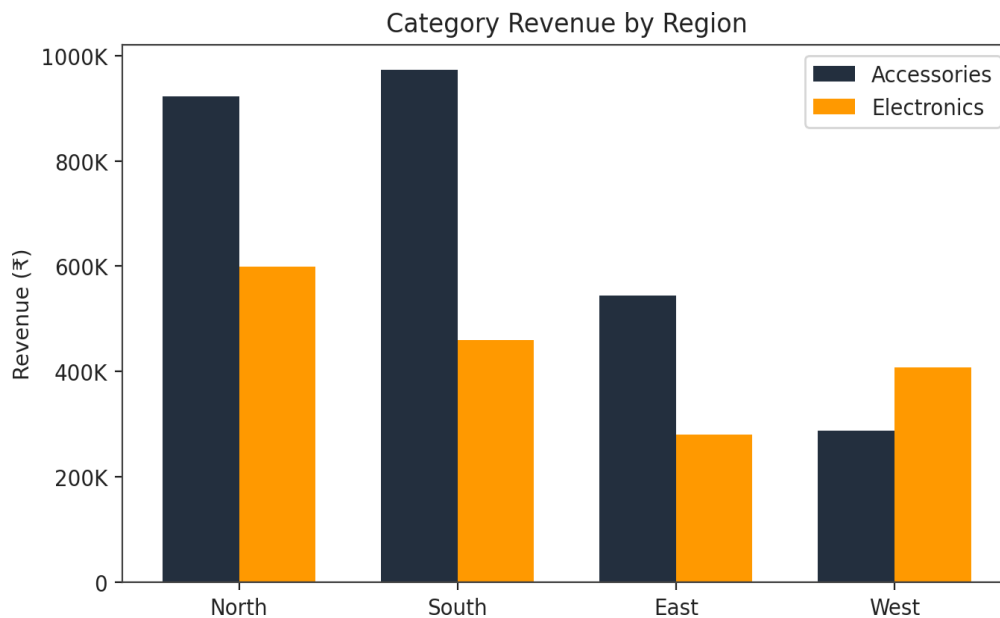
KPI — Most common order size: 4 units — 16 of 60 orders (27%)

Objective 8: Category Revenue Split by Region

Accessories outsells Electronics in every region except West.

Breaking revenue down by both region and category shows the category mix is not uniform across geographies. **Accessories out-earns Electronics in North, South, and East**, while **West is the only region where Electronics revenue exceeds Accessories** (Rs 407,622 vs. Rs 287,816) -- the reverse of the pattern everywhere else.

This kind of region x category crosstab is what separates a one-size-fits-all merchandising strategy from localized assortment and promotion planning -- West may warrant a different inventory mix than the other three regions.



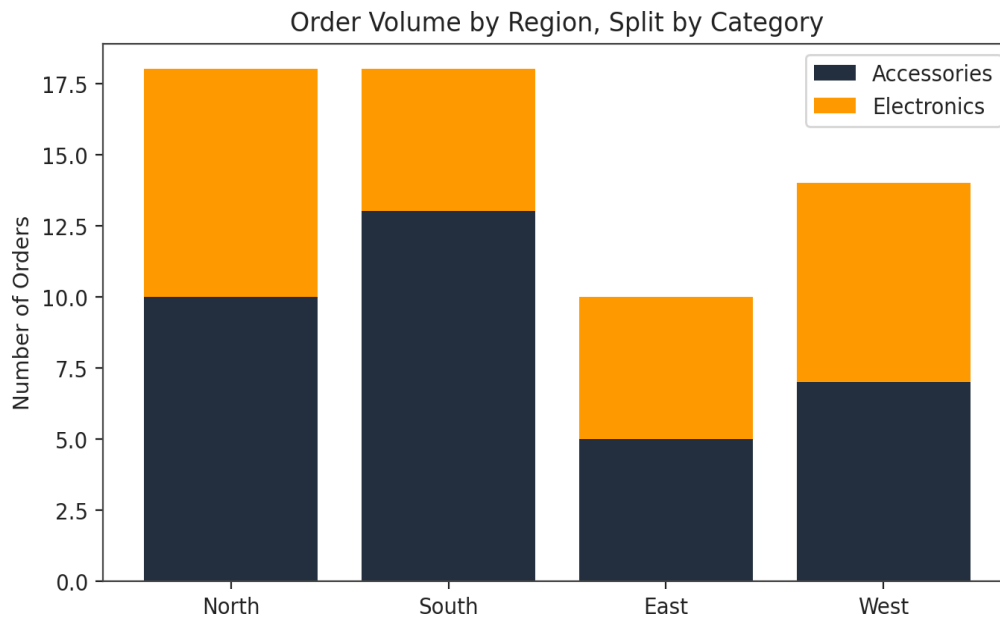
KPI — Region where pattern reverses: West: Electronics Rs 407,622 vs Accessories Rs 287,816

Objective 9: Order Volume by Region and Category

North and South generated 18 and 18 orders respectively -- the two busiest regions by order count.

Order *count* tells a complementary story to order *revenue* (Objective 4). North and South lead on both dimensions, but East shows a meaningfully lower share of Electronics orders (5 of 10) relative to its Accessories volume (5) -- a wider gap proportionally than North or South.

Tracking both order count and revenue by region matters because a region can look healthy on revenue while quietly relying on a handful of large orders -- a pattern that order-count views expose but revenue-only views can mask.



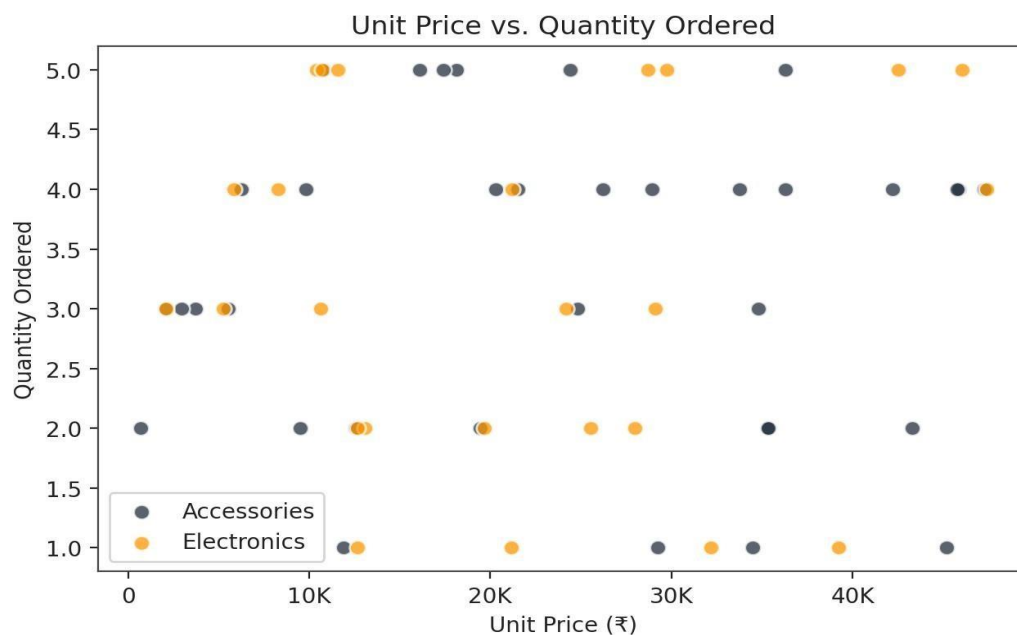
KPI — Busiest regions by order count: North & South — 18 orders each

Objective 10: Relationship Between Unit Price and Quantity Ordered

Correlation between price and quantity: 0.02 -- effectively no linear relationship.

Plotting unit price against quantity ordered shows a correlation coefficient of only **0.02**, indicating price is **not a meaningful driver of order size** in this dataset -- customers buy similar quantities regardless of whether the unit price is low or high, and both categories are spread across the full price range.

The single largest order by value was **Order ORD023** -- 5 units of Keyboard at Rs 46,042 each (Rs 230,210 total), placed in the South region. Because price and quantity don't move together, high-value orders in this dataset come from either high price *or* high quantity, not a systematic combination of both -- worth confirming against a larger sample before using it to inform bundle-pricing strategy.



KPI — Price–quantity correlation: $r = 0.02$ (negligible)

2. Summary of Key Metrics

Metric	Value
Total orders analyzed	60
Total units sold	194
Total revenue	Rs 4,474,143
Average order value (AOV)	Rs 74,569
Average unit price	Rs 22,936
Best-selling product (units)	Mouse (46 units)
Top revenue product	Mouse (Rs 1,191,928)
Leading region (revenue)	North (Rs 1,521,630)
Highest-AOV region	North (Rs 84,535)
Price-quantity correlation	0.02 (negligible)
Products tracked	8
Regions tracked	4

Revenue by Region -- Full Breakdown

Region	Orders	Units	Revenue	AOV
North	18	59	Rs 1,521,630	Rs 84,535
South	18	60	Rs 1,432,154	Rs 79,564
East	10	32	Rs 824,921	Rs 82,492
West	14	43	Rs 695,438	Rs 49,674

3. Product Leaderboard -- Full Detail

Complete per-product breakdown combining every metric referenced across Objectives 1, 2, and 6, sorted by total revenue. This is the table a merchandising review would work from directly.

Product	Orders	Units Sold	Avg. Price	Revenue
Mouse	12	46	Rs 26,591	Rs 1,191,928
Mobile	8	27	Rs 30,672	Rs 874,045
Monitor	10	34	Rs 19,639	Rs 633,562
Keyboard	7	25	Rs 24,775	Rs 606,971
Laptop	8	22	Rs 17,991	Rs 364,238
Tablet	5	17	Rs 20,117	Rs 336,898
Headphones	6	13	Rs 19,585	Rs 237,069
Smartwatch	4	10	Rs 19,972	Rs 229,432

Reading this table: products are ranked by revenue, not units -- compare against the units-sold column to spot products that sell well at volume (e.g. Mouse) versus products that earn well per unit but move less frequently (e.g. Mobile).

Key Findings at a Glance

Finding	Detail
Top product by units	Mouse (46 units)
Top product by revenue	Mouse (Rs 1,191,928)
Leading category	Accessories (61% of revenue)
Leading region	North (Rs 1,521,630)
Region needing attention	West (lowest revenue)
Price-quantity relationship	None detected in current price range

How to read this report: Sections 1-2 set up scope and data structure. Objectives 1-10 each pair one chart with a short, decision-oriented narrative. Sections 3-8 roll the objectives up into summary tables, production-scale context, limitations, and recommended next steps. An appendix with the full order-level dataset follows the conclusion.

4. Next Steps to Move from Descriptive to Predictive

This report answers *what happened*. The natural next layer, if useful:

- **Demand forecasting** per product/region to set reorder points automatically
- **Price elasticity modeling** -- this report found no price-quantity correlation at the current price range, but a wider price-test range could reveal one
- **Regional assortment optimization** -- using the West category-mix anomaly (Objective 8) as a starting hypothesis
- **Cohort and retention analysis** -- once a customer identifier and timestamp are available
- **Anomaly detection** on sudden demand or AOV shifts once the data is refreshed continuously rather than as a static extract

Glossary of Metrics

KPI definitions used consistently throughout this report:

Term		Definition
Units sold	Sum of th	Quantity field across all orders in the relevant grouping (product, region, etc.)
Revenue		Sum of Total (Price x Quantity) across all orders in the relevant grouping
Average Order Value (AOV)		Mean of Total across all orders in the relevant grouping
Average unit price	Me	n of Price across all orders in the relevant grouping (not weighted by quantity)
Order count		Count of distinct Order IDs in the relevant grouping
Correlation coefficient; r	and Coefficient of determination (r ²)	Relationship to +1 (perfect direct relationship), with 0 indicating no linear relationship

A note on currency: Price and Total values in the source file carry no explicit currency label. This report presents them as Indian Rupees (Rs), consistent with typical Amazon-dataset conventions; if the source values are in a different currency, all monetary figures in this report should be re-labeled accordingly -- the underlying relationships and rankings are unaffected either way.

Appendix: Full Order-Level Dataset

The complete 60-row dataset underlying every chart and table in this report, for reference and audit purposes.

Order ID	Product	Category	Price	Qty	Region	Total
ORD001	Mouse	Accessories	18,151	5	South	90,755
ORD002	Mobile	Electronics	21,159	1	North	21,159
ORD003	Mouse	Accessories	29,265	1	South	29,265
ORD004	Mouse	Accessories	6,275	4	South	25,100
ORD005	Mouse	Electronics	10,395	5	West	51,975
ORD006	Tablet	Electronics	5,861	4	East	23,444
ORD007	Mouse	Accessories	3,743	3	South	11,229
ORD008	Headphones	Accessories	45,191	1	West	45,191
ORD009	Mobile	Accessories	45,857	4	North	183,428
ORD010	Tablet	Accessories	10,820	5	South	54,100
ORD011	Headphones	Accessories	682	2	South	1,364
ORD012	Laptop	Accessories	5,537	3	West	16,611
ORD013	Laptop	Electronics	13,067	2	West	26,134
ORD014	Monitor	Accessories	35,293	2	East	70,586
ORD015	Headphones	Electronics	27,974	2	East	55,948
ORD016	Mobile	Accessories	36,299	4	East	145,196
ORD017	Mobile	Accessories	2,038	3	West	6,114
ORD018	Monitor	Accessories	24,405	5	East	122,025
ORD019	Monitor	Electronics	29,753	5	North	148,765
ORD020	Laptop	Electronics	24,188	3	South	72,564
ORD021	Mouse	Electronics	21,187	4	North	84,748
ORD022	Mouse	Electronics	47,390	4	North	189,560
ORD023	Keyboard	Electronics	46,042	5	South	230,210
ORD024	Laptop	Accessories	43,311	2	South	86,622
ORD025	Mouse	Accessories	17,396	5	West	86,980
ORD026	Tablet	Electronics	25,545	2	North	51,090
ORD027	Keyboard	Accessories	16,113	5	North	80,565
ORD028	Headphones	Accessories	28,916	4	South	115,664
ORD029	Smartwatch	Accessories	11,899	1	North	11,899
ORD030	Smartwatch	Electronics	12,659	2	East	25,318
ORD031	Mobile	Accessories	35,330	2	South	70,660
ORD032	Keyboard	Accessories	34,483	1	South	34,483
ORD033	Monitor	Electronics	32,181	1	East	32,181
ORD034	Mobile	Accessories	42,208	4	North	168,832
ORD035	Keyboard	Accessories	34,798	3	North	104,394

Order ID	Product	Category	Price	Qty	Region	Total
ORD036	Mobile	Accessories	33,774	4	North	135,096
ORD037	Laptop	Electronics	8,271	4	West	33,084
ORD038	Keyboard	Electronics	10,633	3	West	31,899
ORD039	Smartwatch	Electronics	29,105	3	South	87,315
ORD040	Monitor	Accessories	2,957	3	North	8,871
ORD041	Mobile	Electronics	28,712	5	East	143,560
ORD042	Monitor	Accessories	20,281	4	South	81,124
ORD043	Keyboard	Accessories	21,518	4	North	86,072
ORD044	Laptop	Electronics	5,281	3	South	15,843
ORD045	Smartwatch	Accessories	26,225	4	North	104,900
ORD046	Mouse	Electronics	39,211	1	North	39,211
ORD047	Mouse	Electronics	42,506	5	West	212,530
ORD048	Headphones	Electronics	2,078	3	North	6,234
ORD049	Laptop	Accessories	24,838	3	West	74,514
ORD050	Laptop	Accessories	19,433	2	North	38,866
ORD051	Tablet	Accessories	12,584	2	East	25,168
ORD052	Monitor	Electronics	10,736	5	South	53,680
ORD053	Mouse	Accessories	47,270	4	South	189,080
ORD054	Mouse	Accessories	36,299	5	East	181,495
ORD055	Monitor	Accessories	9,529	2	West	19,058
ORD056	Tablet	Accessories	45,774	4	South	183,096
ORD057	Monitor	Electronics	19,666	2	West	39,332
ORD058	Monitor	Electronics	11,588	5	North	57,940
ORD059	Keyboard	Accessories	9,837	4	West	39,348
ORD060	Headphones	Electronics	12,668	1	West	12,668

Dataset Overview

Each of the 60 records represents a single completed order with the fields below. No date/timestamp field is present in this dataset, so trend-over-time analysis is out of scope; instead, this report focuses on cross-sectional patterns -- product, category, region, price and quantity -- which is where the dataset carries signal.

Field	Type	Description
Order ID	Text	Unique identifier per order (ORD001-ORD060)
Product	Text	8 distinct products (e.g. Mouse, Laptop, Smartwatch)
Category	Text	Accessories or Electronics
Price	Numeric (Rs)	Unit price at time of order
Quantity	Numeric	Units purchased in the order (1-5)
Region	Text	North, South, East, or West
Total	Numeric (Rs)	Price x Quantity

End of report.